

Logistic Regression & Gaussian Discriminant Analysis

A Complete Worked Example: University Admissions

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1 The Problem and Dataset

Task. Predict whether a student will be *admitted* ($y = 1$) or *rejected* ($y = 0$) to a university programme, based on two exam scores.

Feature Vector

For each student i :

$$\mathbf{x}^{(i)} = \begin{pmatrix} x_1^{(i)} \\ x_2^{(i)} \end{pmatrix} \in \mathbb{R}^2, \quad x_1 = \text{Exam 1 score}, \quad x_2 = \text{Exam 2 score}, \quad y \in \{0, 1\}.$$

1.1 Training Data

We have $N = 5$ training examples.

Student	Exam 1 (x_1)	Exam 2 (x_2)	Admitted? (y)
1	45	85	1 (Yes)
2	85	46	1 (Yes)
3	60	83	1 (Yes)
4	30	49	0 (No)
5	68	28	0 (No)

Class summary: **3 admitted** ($y = 1$: students 1, 2, 3) and **2 rejected** ($y = 0$: students 4, 5).

1.2 Test Point

After fitting both models we will classify:

$$\mathbf{x}^* = \begin{pmatrix} 45 \\ 85 \end{pmatrix} \quad (\text{Exam 1} = 45, \text{ Exam 2} = 85).$$

2 Logistic Regression

2.1 Model

Logistic regression is a *discriminative* model: it directly models the conditional probability

$$P(y = 1 \mid \mathbf{x}; \boldsymbol{\theta}) = h_{\boldsymbol{\theta}}(\mathbf{x}) = \sigma(\boldsymbol{\theta}^\top \tilde{\mathbf{x}}) = \frac{1}{1 + e^{-\boldsymbol{\theta}^\top \tilde{\mathbf{x}}}},$$

where $\tilde{\mathbf{x}} = (1, x_1, x_2)^\top$ is the augmented feature vector (with a bias term) and $\boldsymbol{\theta} = (\theta_0, \theta_1, \theta_2)^\top \in \mathbb{R}^3$.

The **decision rule** is: predict $\hat{y} = 1$ if $h_{\boldsymbol{\theta}}(\mathbf{x}) \geq 0.5$, i.e. $\boldsymbol{\theta}^\top \tilde{\mathbf{x}} \geq 0$.

2.2 Cost Function (Cross-Entropy Loss)

Applying Maximum Likelihood Estimation under a Bernoulli model gives the *cross-entropy* (log-loss) cost:

$$J(\boldsymbol{\theta}) = -\frac{1}{N} \sum_{i=1}^N \left[y^{(i)} \log h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) + (1 - y^{(i)}) \log(1 - h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)})) \right].$$

J is convex in $\boldsymbol{\theta}$, so gradient descent is guaranteed to find the global minimum.

2.3 Gradient Descent Update Rule

$$\theta_j \leftarrow \theta_j - \frac{\alpha}{N} \sum_{i=1}^N (h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) - y^{(i)}) x_j^{(i)}, \quad j = 0, 1, 2.$$

2.4 Iteration 0 — Initial State

Iteration 0: $\boldsymbol{\theta} = (0, 0, 0)^\top$

With all weights zero, $\boldsymbol{\theta}^\top \tilde{\mathbf{x}}^{(i)} = 0$ for every example, so

$$h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) = \sigma(0) = \frac{1}{2} \quad \forall i.$$

The initial cost is

$$J(\boldsymbol{\theta}) = -\frac{1}{N} \sum_{i=1}^N \left[y^{(i)} \log \frac{1}{2} + (1 - y^{(i)}) \log \frac{1}{2} \right] = -\log \frac{1}{2} = \ln 2 \approx \mathbf{0.6931}.$$

2.5 Iteration 1 — First Gradient Step

Iteration 1: Computing the gradient at $\boldsymbol{\theta} = \mathbf{0}$

The residuals $h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) - y^{(i)}$ at initialisation are:

i	$h_{\theta}(\mathbf{x}^{(i)})$	$y^{(i)}$	$h - y$
1	0.5	1	-0.5
2	0.5	1	-0.5
3	0.5	1	-0.5
4	0.5	0	+0.5
5	0.5	0	+0.5

Gradient components (using learning rate $\alpha = 0.0001$):

$$\frac{\partial J}{\partial \theta_0} = \frac{1}{N} \sum_i (h_i - y^{(i)}) \cdot 1 = \frac{-0.5 - 0.5 - 0.5 + 0.5 + 0.5}{5} = -0.1000,$$

$$\begin{aligned} \frac{\partial J}{\partial \theta_1} &= \frac{1}{N} \sum_i (h_i - y^{(i)}) \cdot x_1^{(i)} \\ &= \frac{(-0.5)(45) + (-0.5)(85) + (-0.5)(60) + (0.5)(30) + (0.5)(68)}{5} = \frac{-46}{5} = -9.2, \end{aligned}$$

$$\begin{aligned} \frac{\partial J}{\partial \theta_2} &= \frac{1}{N} \sum_i (h_i - y^{(i)}) \cdot x_2^{(i)} \\ &= \frac{(-0.5)(85) + (-0.5)(46) + (-0.5)(83) + (0.5)(49) + (0.5)(28)}{5} = \frac{-68.5}{5} = -13.7. \end{aligned}$$

Updated parameters:

$$\begin{aligned} \theta_0 &\leftarrow 0 - 0.0001 \times (-0.1) = 0.0000, \\ \theta_1 &\leftarrow 0 - 0.0001 \times (-9.2) = 0.0009, \\ \theta_2 &\leftarrow 0 - 0.0001 \times (-13.7) = 0.0014. \end{aligned}$$

New cost: $J \approx 0.6682$.

2.6 Gradient Descent Progress

The table below summarises the first seven iterations.

Iter.	θ_0	θ_1	θ_2	$J(\theta)$
0	0.0000	0.0000	0.0000	0.6931
1	0.0000	0.0009	0.0014	0.6682
2	0.0000	0.0016	0.0025	0.6510
3	0.0000	0.0022	0.0035	0.6389
4	0.0000	0.0027	0.0044	0.6304
5	0.0000	0.0030	0.0051	0.6243
6	0.0000	0.0032	0.0058	0.6198
7	0.0000	0.0034	0.0063	0.6164

2.7 Decision Boundary (Converged Model)

The converged parameters are $\hat{\theta} = (-25.16, 0.206, 0.201)^\top$.

2.8 Prediction for $\mathbf{x}^* = (45, 85)^\top$

$$\begin{aligned} z &= \hat{\theta}_0 + \hat{\theta}_1 \cdot 45 + \hat{\theta}_2 \cdot 85 \\ &= -25.16 + 0.206 \times 45 + 0.201 \times 85 = -25.16 + 9.27 + 17.09 = 1.195. \end{aligned}$$

$$P(y = 1 \mid \mathbf{x}^*) = \sigma(1.195) = \frac{1}{1 + e^{-1.195}} \approx 0.768.$$

Since $0.768 \geq 0.5$: $\hat{y} = 1$ (**Admitted**).